

In [10]: # Q1. Write a Python program to input two numbers and display their sum, difference, product and quotient.

```
a=int(input("enter first number :-"))
b=int(input("enter second number :- "))

_sum=a+b
_difference=a-b
_product=a*b
_quotient=a/b

print("sum =", _sum, "\ndifference =", _difference, "\nproduct =",
_product, "\nquotient =", _quotient)
```

```
enter first number :- 2
enter second number :- 2
```

```
sum = 4
difference = 0
product = 4
quotient = 1.0
```

In [11]: # Q2. Write a Python program to input the length and breadth of a rectangle and calculate its area and perimeter.

```
l=float(input("enter length of rectange :-"))
b=float(input("enter breadth of rectange :-"))

a=l*b
p=2*(l+b)

print("area = ",a,"\nperemeter = ",p)
```

```
enter length of rectange :- 2
enter breadth of rectange :- 3
```

```
area = 6.0
peremeter = 10.0
```

In [12]: #Q3. Write a Python program to input the radius of a circle and calculate its area and circumference. Use 3.14 as the value of π .

```
r=float(input("enter the radius"))

a=3.14*4**2
c=2*3.14*r

print("area = ",a,"\ncircumference = ",c)
```

```
enter the radius 4
```

```
area = 50.24
circumference = 25.12
```

In [15]: #Q4. Write a Python program to input principal amount, rate of interest and time and calculate the Simple Interest and Total Amount.

```
P=float(input("enter principal amount"))
R=float(input("enter rate of interest"))
T=float(input("enter time period"))

SI=P*R*T/100
Amount=P+SI

print("simple interest =",SI,"\nAmount = ",Amount)
```

```
enter principal amount 1000
enter rate of interest 8
enter time period 5
```

```
simple interest = 400.0
Amount = 1400.0
```

In [16]: #Q5. Write a Python program to input the marks obtained in five subjects and calculate the total marks and percentage.

```
p=float(input("enter marks obtained in physics :- "))
c=float(input("enter marks obtained in chemistry :- "))
m=float(input("enter marks obtained in maths:- "))
cs=float(input("enter marks obtained in computer science :- "))
e=float(input("enter marks obtained in english :- "))

t=p+c+m+cs+e

p=t/400*100

print("total marks = ",t,"\npercentage = ",p)
```

```
enter marks obtained in physics :- 80
enter marks obtained in chemistry :- 80
enter marks obtained in maths:- 80
enter marks obtained in computer science :- 80
enter marks obtained in english :- 80

total marks = 400.0
percentage = 100.0
```

In [17]: # Q6. Write a Python program to input a temperature in Celsius and convert it into Fahrenheit. $F=9*C/5 + 32$

```
C=float(input("enter tempture in Celsius"))

F=9*C/5 + 32

print("Fahrenheit = ",F)
```

```
enter tempture in Celsius 45

Fahrenheit = 113.0
```

In [18]: #Q7. Write a Python program to input a number and display its square, cube and fourth power.

```
a=int(input("enter your number"))

print("square =",a**2,"\ncube",a**3,"\nfourth power",a**4)
```

```
enter your number 2

square = 4
cube 8
fourth power 16
```

```
In [19]: #Q8. Write a Python program to input two numbers and swap their values
         using Python's multiple assignment statement.
         a=int(input("enter 1 number"))
         b=int(input("enter 2 number"))

         print("before swap",a,b)

         a,b=b,a

         print("after swap",a,b)
```

```
enter 1 number 2
enter 2 number 3

before swap 2 3
after swap 3 2
```

```
In [20]: '''
         Q9. Write a Python program to input a distance in kilometres and convert
         it into miles.
         Use:                                1 km=0.621371 miles
         '''

         a=float(input("enter distance in km"))

         s=a*0.621371

         print("miles = ",s)
```

```
enter distance in km 5

miles =  3.106855
```

```
In [21]: #Q10. Write a Python program to input the base and height of a triangle
         and calculate its area.      Area=1/2×Base×Height

         b=float(input("enter base of trinage"))
         h=float(input("enter height of triangle"))

         a=1/2*b*h

         print(a)
```

```
enter base of trinage 2
enter height of triangle 2

2.0
```

In [22]: #Q11. Write a Python program to input a number and check whether it is positive, negative or zero.

```
a=int(input("enter your number :- "))
if a>0:
    print("positive")
elif a<0:
    print("negative")
else:
    print("zero")
```

enter your number :- 3

positive

In [23]: #Q12. Write a Python program to input a number and check whether it is even or odd.

```
a=int(input("enter your number :- "))
if a%2==0:
    print("even")
else:
    print("odd")
```

enter your number :- 7

odd

In []: #Q13. Write a Python program to input three numbers and display the largest number.

```
a=int(input("enter first number :- "))
b=int(input("enter second number :- "))
c=int(input("enter third number :- "))

if a>b and a>c:
    print("largest number =",a)
elif b>a and b>c:
    print("largest number =",b)
else:
    print("largest number =",c)
```

In [35]: #Q14. Write a Python program to input a student's marks and display the grade according to the following criteria.

```
marks=float(input("enter your marks :- "))

if marks>=90:
    print("Grade A")
elif marks>=75:
    print("Grade B")
elif marks>=50:
    print("Grade C")
else:
    print("Grade D")
```

enter your marks :- 80

Grade B

In [25]: #Q15. Write a Python program to input a number and check whether it is divisible by both 3 and 5.

```
a=int(input("enter your number :- "))

if a%3==0 and a%5==0:
    print("divisible by both 3 and 5")
else:
    print("not divisible by both 3 and 5")
```

enter your number :- 15

divisible by both 3 and 5

In [26]: # Q16. Write a Python program using a for loop to display the numbers from 1 to 20.

```
for i in range(1,21):  
    print(i)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20
```

In [28]: #Q17. Write a Python program to input a number and display its multiplication table from 1 to 10.

```
a=int(input("enter your number"))  
  
for i in range(1,11):  
    print(a,"x",i,"=",a*i)
```

```
enter your number 5
```

```
5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30  
5 x 7 = 35  
5 x 8 = 40  
5 x 9 = 45  
5 x 10 = 50
```

In [29]: #Q18. Write a Python program using a while loop to calculate the sum of numbers from 1 to 10.

```
i=1
s=0
while i<=10:
    s=s+i
    i=i+1

print("sum =",s)
```

sum = 55

In [31]: #Q19. Write a Python program to input a positive integer and calculate the sum of its digits.

```
a=int(input("enter a positive number :- "))
s=0

while a>0:
    d=a%10
    s=s+d
    a=a//10

print("sum of digits =",s)
```

enter a positive number :- 54

sum of digits = 9

In [34]: #Q20. Write a Python program to input a number and check whether it is a prime number or not.

```
a=int(input("enter your number :- "))
c=0
for i in range(1,a+1):
    if a%i==0:
        c=c+1
if c==2:
    print("prime number")
else:
    print("not a prime number")
```

enter your number :- 8

not a prime number